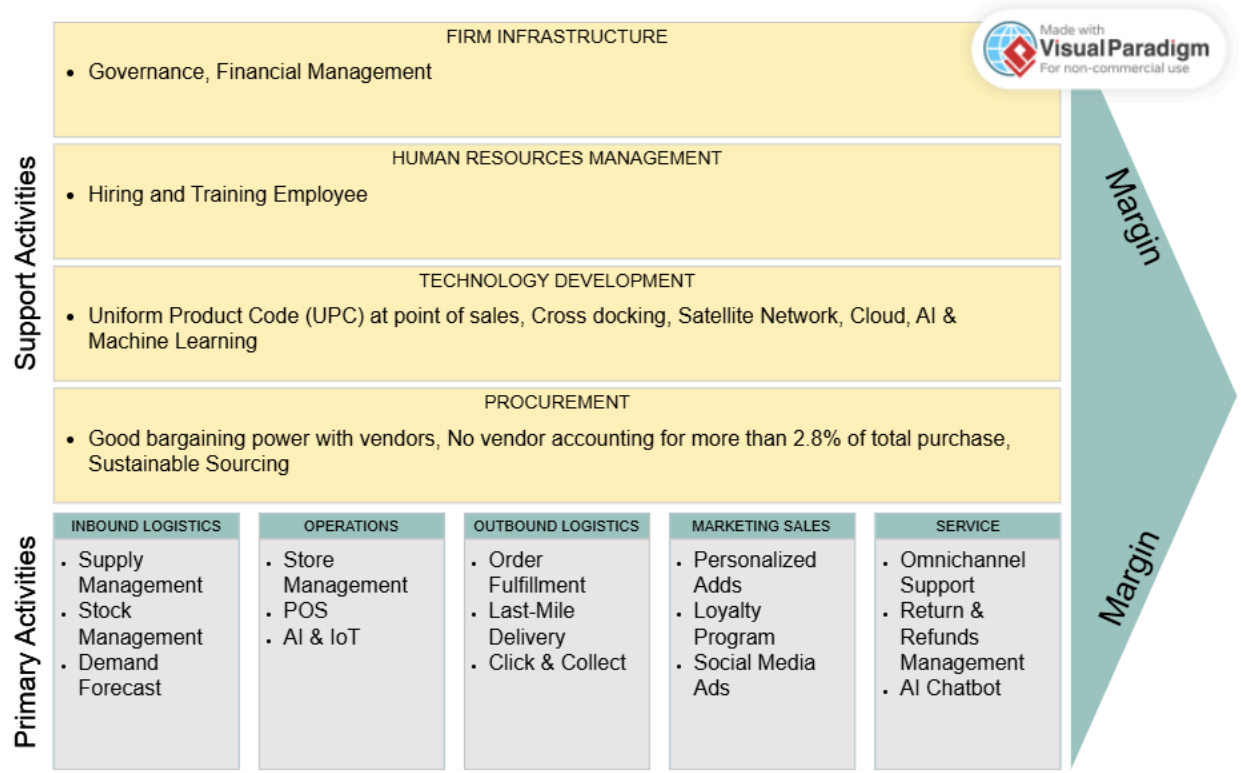
Enterprise Architecture - TOGAF

FMCG Retail Industry

Architecture Vision



1. Introduction

1.1 Objective of the Document

Dokumen ini menjelaskan **Architecture Vision** untuk perusahaan retail berdasarkan framework **TOGAF**. Vision ini mendukung transformasi digital dengan fokus pada **omnichannel retailing, data-driven decision making, serta efisiensi operasional** melalui integrasi teknologi modern.

1.2 Scope

Lingkup arsitektur ini mencakup seluruh **rantai nilai bisnis retail**, mulai dari **manajemen stok, pengelolaan pelanggan, transaksi penjualan, hingga supply chain & logistik** dengan pendekatan **omnichannel**.

1.3 Problem Statement

Industri ritel menghadapi tantangan yang signifikan dalam memberikan **seamless customer experience**, **mengoptimalkan supply chain operations, dan mengadopsi digital transformation strategies.** Model bisnis tradisional kesulitan untuk mengikuti **perkembangan ekspektasi konsumen, meningkatnya persaingan dari platform e-commerce, dan permintaan untuk data-driven decision-making**. Peritel harus memodernisasi arsitektur bisnis dan TI mereka agar tetap kompetitif dan efisien.

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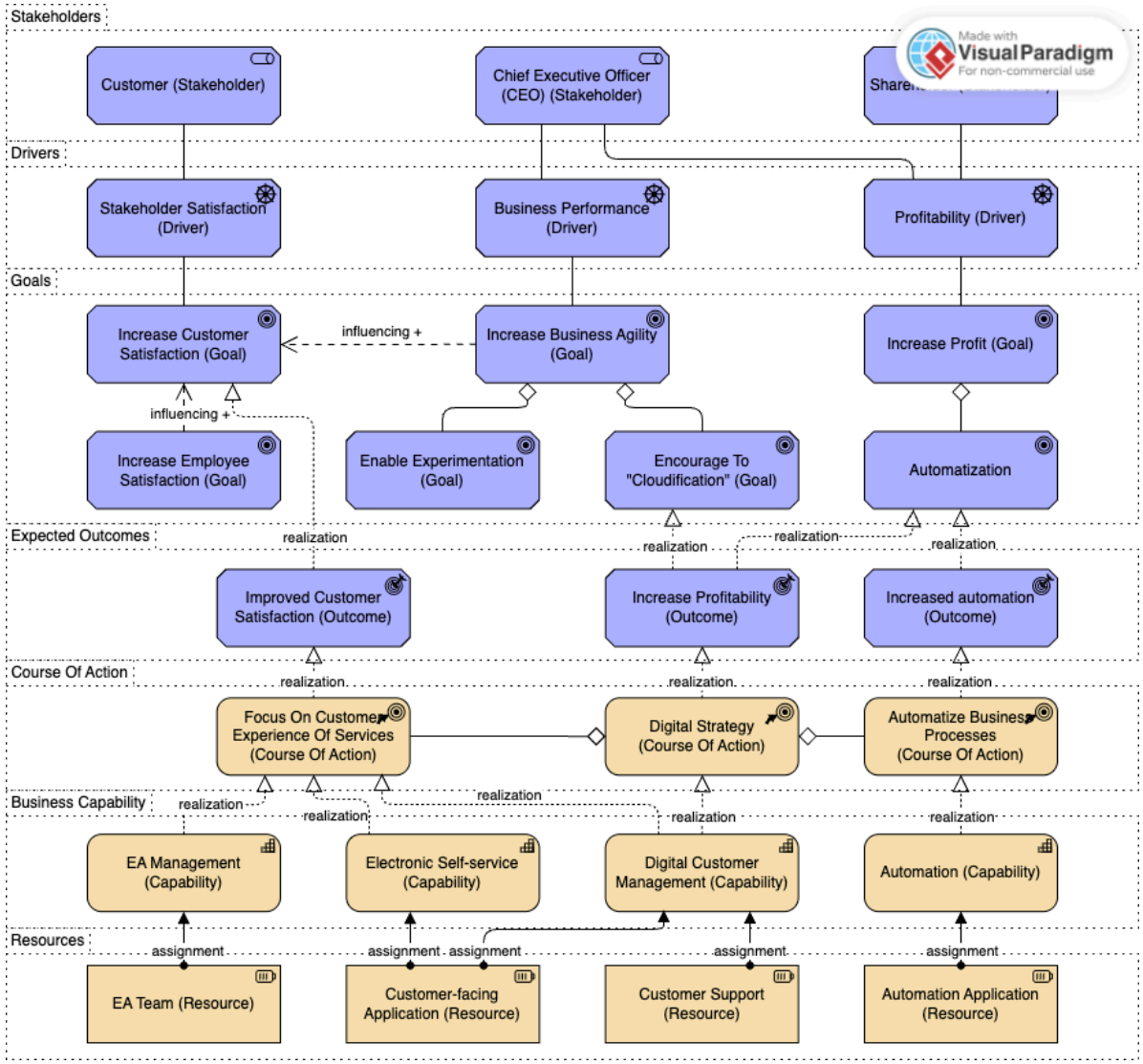
2. Architecture Vision Statement

"To build a modern, scalable, and customer-focused retail enterprise architecture that enables seamless omnichannel experiences, data-driven insights, and efficient supply chain management while ensuring security and compliance."



3. Business Drivers & Stakeholders

Key Business Drivers:

- Evolving customer expectations for omnichannel experiences
- Competition from e-commerce platforms
- Need for data-driven decision-making
- Operational efficiency and cost reduction
- Compliance with retail regulations and data privacy laws

Key Stakeholders:

Stakeholder Group	Concerns & Interests
CEO / Business Leaders	Revenue growth, brand positioning
CIO / IT Managers	Technology adoption, system scalability
Marketing Team	Customer insights, personalized engagement
Store Operations	Inventory management, POS efficiency
Supply Chain Team	Logistics, vendor integration
Customers	Seamless shopping experience, data security

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4. Solution Concept

The retail architecture will consist of:

- Omnichannel Retail Platform: Unified online & offline experience
- Cloud-based Infrastructure: Scalability and flexibility

- AI-driven Analytics: Customer insights and demand forecasting
- ERP & CRM Integration: Enhanced supply chain & customer relationship management
- Cybersecurity Framework: Data protection and fraud prevention

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5. High-Level Architecture Principles

Principle	Description
Customer-Centricity	Prioritize seamless, personalized experiences
Data-Driven Decision Making	Leverage AI & analytics for insights
Scalability & Agility	Use cloud and microservices for flexibility
Security & Compliance	Ensure GDPR, PCI DSS, and other standards
Integration & Interoperability	API-driven architecture for seamless integration

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6. Benefits & Outcomes

✔️ Enhanced customer experience through personalization and omnichannel integration
✔️ Increased operational efficiency with automation and AI-driven analytics
✔️ Improved inventory and supply chain management
✔️ Scalable, future-proof IT architecture for business growth

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7. Next Steps

- Conduct stakeholder workshops for requirement gathering
- Define high-level business and IT requirements
- Develop a Capability Map and Transition Roadmap

Business Architecture

1. Purpose & Objective

The Business Architecture aims to create a structured and scalable retail framework that supports omnichannel customer experiences, enhances operational efficiency, and enables data-driven decision-making.

Key Business Goals:

- **Deliver seamless omnichannel experiences** (online, in-store, mobile, social media).
- **Optimize supply chain and inventory management** for real-time responsiveness.
- **Enhance customer engagement** with AI-driven personalization.
- **Improve operational efficiency** by integrating core business functions.
- **Ensure regulatory compliance** with data privacy and payment security standards.

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2. Business Architecture Framework

2.1 Business Capabilities

Business capabilities represent the fundamental functions needed to achieve retail business objectives.

Capability Domain	Key Capabilities
Customer Engagement	Omnichannel experience, loyalty programs, customer support
Sales & Marketing	Pricing strategy, promotions, digital marketing
Merchandising & Inventory	Demand forecasting, product assortment, inventory control

Supply Chain & Logistics	Supplier collaboration, order fulfillment, last-mile delivery
Store Operations	POS systems, workforce management, self-checkout
Finance & Compliance	Billing, fraud detection, tax compliance
Data & Analytics	Business intelligence, AI-driven insights, predictive analytics

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2.2 Business Process Model

The retail industry operates through interconnected business processes. Below is a high-level representation of key processes:

Customer Journey & Sales Process

1. **Customer Engagement** → Personalized marketing campaigns & promotions
2. **Order Placement** → Customers place orders via in-store, web, or mobile
3. **Payment & Billing** → Secure transactions with multiple payment options
4. **Order Fulfillment** → Automated stock allocation & logistics coordination
5. **Delivery & Returns** → Efficient order tracking & hassle-free returns

Supply Chain & Inventory Process

1. **Demand Forecasting** → AI-driven prediction models
2. **Supplier Management** → Vendor onboarding & contract management
3. **Inventory Replenishment** → Automated stock ordering
4. **Warehouse & Logistics** → Fulfillment center optimization
5. **Delivery Optimization** → Last-mile delivery & real-time tracking

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3. Business Roles & Stakeholders

Each business process involves key stakeholders. Below is a high-level mapping:

Business Role	Responsibilities
Retail Store Manager	Oversees in-store operations & customer service
E-commerce Manager	Manages online sales channels & digital platforms
Supply Chain Manager	Optimizes procurement, logistics, and inventory
Marketing Team	Executes branding, promotions & customer engagement
IT Team	Implements retail technology, POS & security solutions
Finance & Compliance	Manages payments, audits & data regulations

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4. Business Information Model

The Business Information Model defines the key data entities that support business processes.

Data Entity	Description
Customer Profile	Personal details, preferences, purchase history
Product Catalog	SKU details, pricing, promotions, availability
Sales Transactions	POS, online, and mobile sales data
Supplier Information	Vendor contracts, shipment schedules, SLAs
Inventory Data	Stock levels, warehouse locations, replenishment schedules

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5. Business Principles & Guidelines

Principle	Description

Customer-Centricity	Ensure a seamless, personalized shopping experience
Data-Driven Decision Making	Use AI & analytics for demand forecasting & marketing
Scalability & Agility	Cloud-based architecture for business growth
Omnichannel Integration	Ensure unified customer experiences across all touchpoints
Regulatory Compliance	Adhere to GDPR, PCI DSS, and other standards

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6. Business Architecture Roadmap

The roadmap outlines key milestones in transforming the retail industry's business architecture.

Phase	Key Initiatives
Phase 1: Digital Enablement	Cloud adoption, e-commerce platform upgrade
Phase 2: AI & Analytics	Customer insights, predictive analytics for inventory
Phase 3: Omnichannel Integration	Unified customer engagement, seamless UX
Phase 4: Automation & Innovation	AI-powered chatbots, smart checkout, automation

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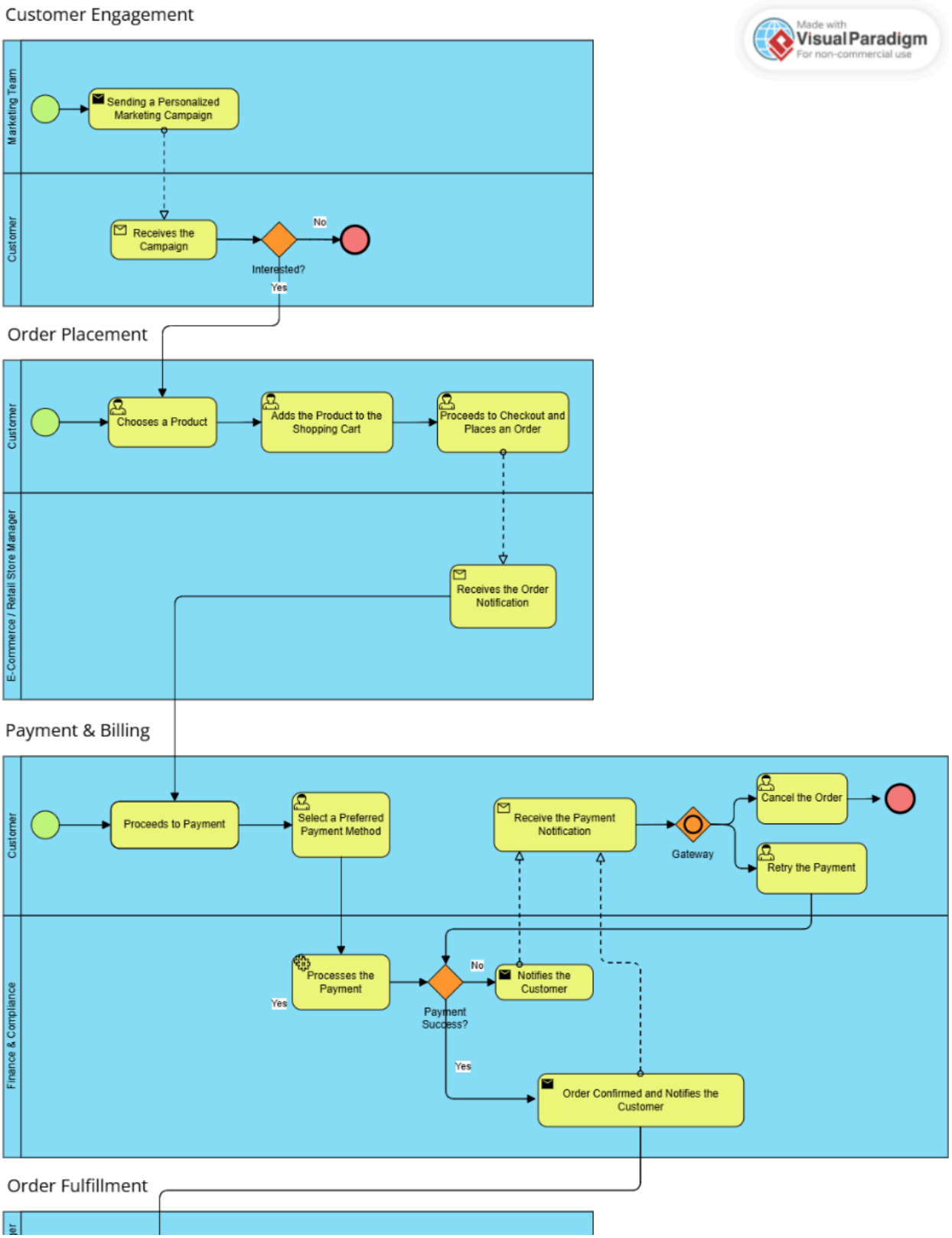
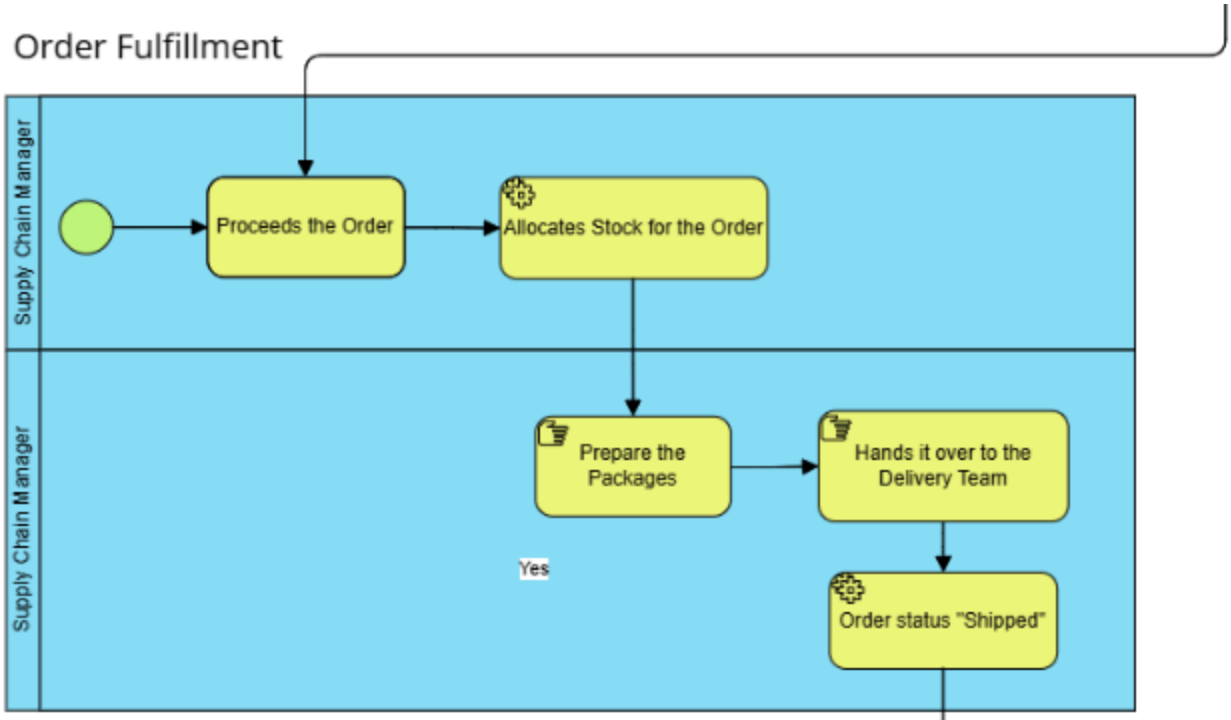
7. Key Outcomes & Benefits

✔️ **Enhanced customer experience** through personalized engagement
✔️ **Optimized inventory & supply chain management** with real-time visibility
✔️ **Seamless omnichannel retail experience** across digital & physical stores
✔️ **Increased operational efficiency** through integrated systems
✔️ **Improved decision-making** with AI-driven analytics

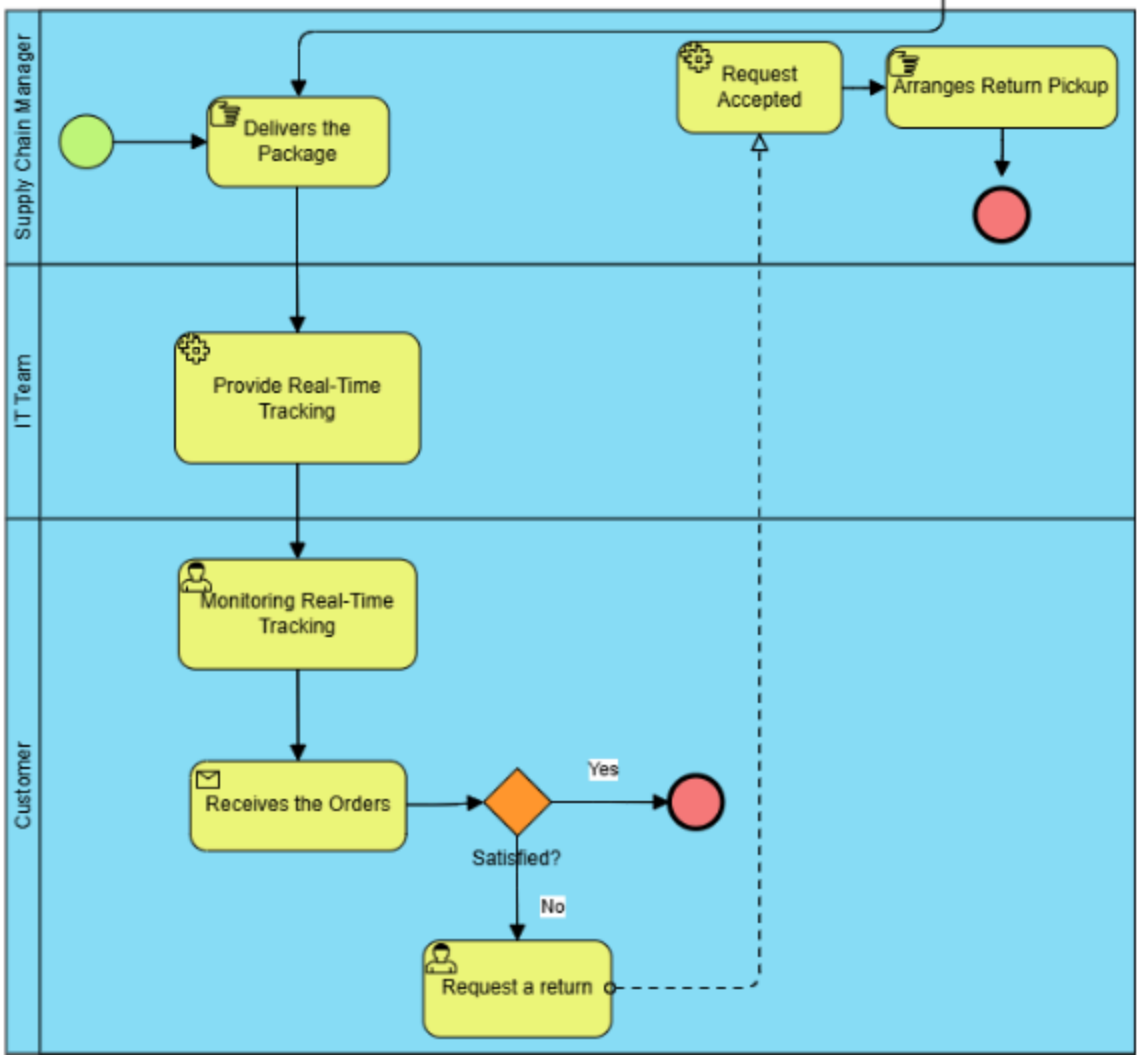
Customer Journey & Sales Process



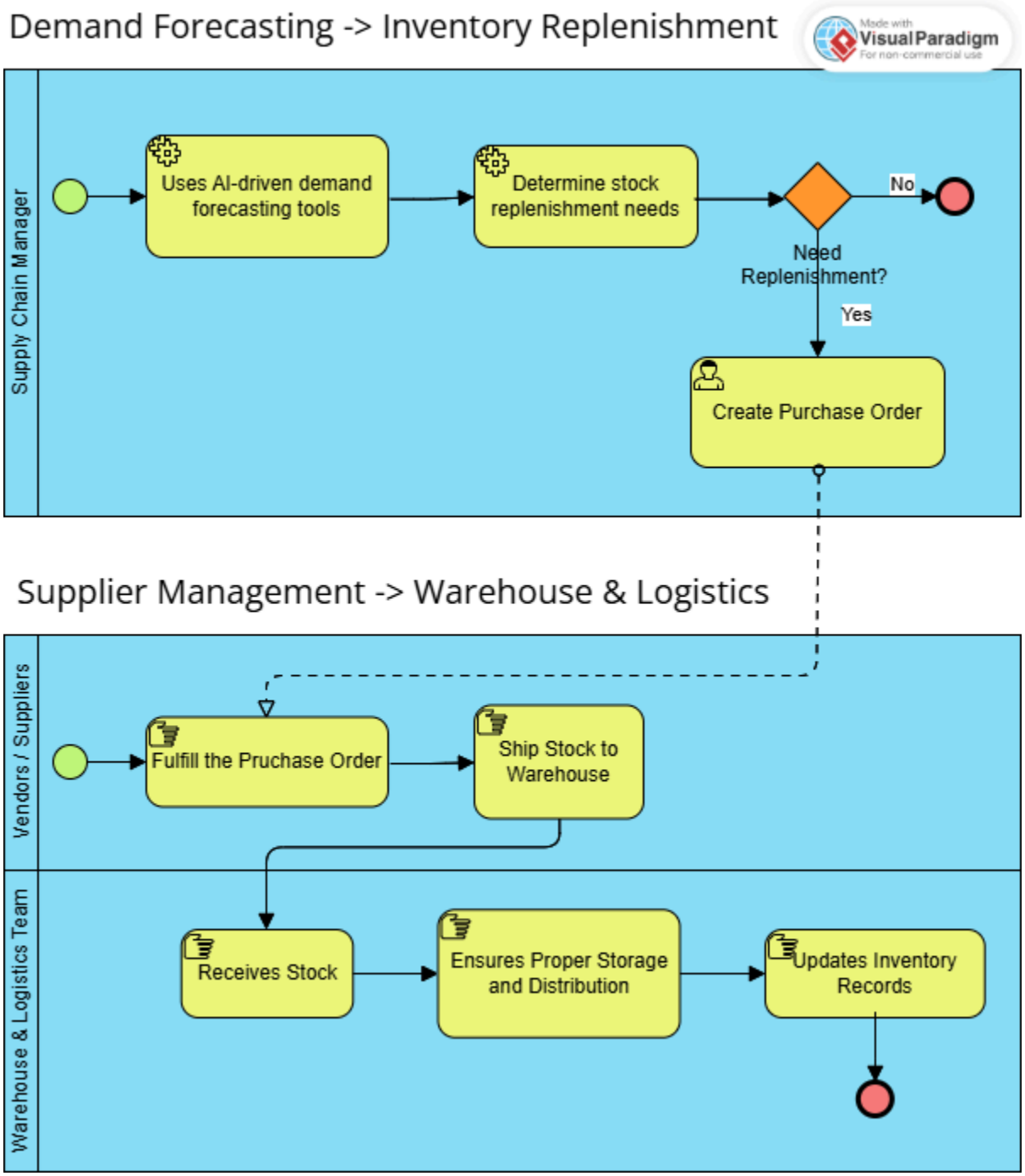
Order Fulfillment



Delivery & Returns - Delivery Optimalization

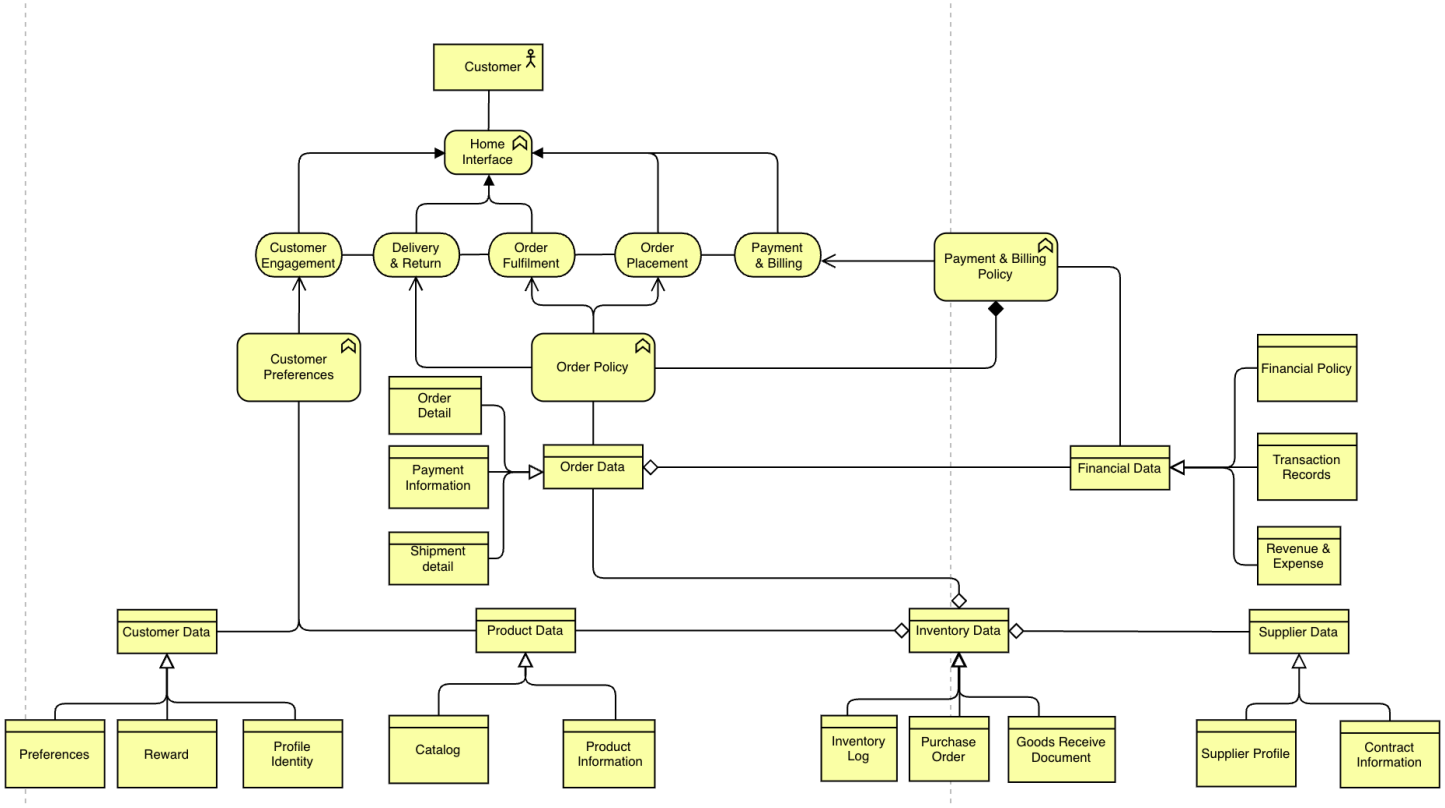


Supply Chain & Inventory Process



Information Systems Architecture

1. Data Architecture



Data Architecture menjelaskan struktur, aliran, dan manajemen data di seluruh sistem perusahaan ritel. Berikut adalah komponen detailnya:

A. Data Entities and Domain Model

1. **Customer Data**
    - **Customer Profile**: Nama, alamat, email, nomor telepon.
    - **Preferences**: Preferensi produk, riwayat pencarian.
    - **Loyalty and Rewards**: Poin, level loyalitas, histori penukaran.
2. **Product Data**
    - **Product Catalog**: Nama produk, deskripsi, kategori, gambar.
    - **Inventory Details**: Stok, lokasi gudang, reorder point.
    - **Pricing Information**: Harga reguler, harga diskon, bundling.

3. **Order Data**
    - **Order Details**: Nomor pesanan, tanggal, status pesanan.
    - **Payment Information**: Metode pembayaran, status pembayaran.
    - **Shipment Details**: Alamat pengiriman, nomor resi, status pengiriman.
4. **Supplier Data**
    - **Supplier Profile**: Nama supplier, kontak, alamat.
    - **Contract Information**: Detail kontrak, harga pembelian, SLA.
5. **Financial Data**
    - **Transaction Records**: Laporan penjualan, pengembalian dana.
    - **Revenue and Expenses**: Pendapatan, biaya operasional, profit margin.

6. Inventory Data

- **Inventory Log :** Laporan log transaksi keluar masuk untuk inventori
- **Purchase Order Document:** Laporan Pemesanan barang
- **Goods Received Document:** Laporan Penerimaan Barang

B. Data Flow and Integration

Data mengalir di antara beberapa sistem dalam perusahaan ritel:

1. **Customer Data Flow**:
    - Dari **Website/Mobile App** ke **Customer Data Platform (CDP)**.
    - Dari **CDP** ke **Marketing Automation** untuk personalisasi.
2. **Product Data Flow**:
    - Dari **Product Information Management (PIM)** ke **E-commerce Platform**.
    - Dari **PIM** ke **Warehouse Management System (WMS)** untuk stok real-time.
3. **Order Data Flow**:
    - Dari **E-commerce Platform** ke **Order Management System (OMS)**.
    - Dari **OMS** ke **Payment Gateway** dan **WMS** untuk fulfillment.
4. **Inventory Data Flow**:
    - Dari **WMS** ke **E-commerce Platform** untuk update stok.
    - Dari **WMS** ke **Supply Chain Management (SCM)** untuk replenishment.

C. Data Storage and Management

1. **Data Warehouse**:
    - Menyimpan data historis penjualan dan inventaris untuk analitik.
    - Menggunakan **BigQuery, Redshift, atau Snowflake**.
2. **Data Lake**:
    - Menampung data tidak terstruktur seperti log aplikasi dan ulasan pelanggan.
    - Menggunakan **AWS S3, Azure Data Lake, atau Google Cloud Storage**.
3. **Operational Databases**:
    - **Relational Database**: MySQL/PostgreSQL untuk data transaksi.
    - **NoSQL Database**: MongoDB/DynamoDB untuk data katalog dan pelanggan.

D. Master Data Management (MDM)

Mengelola data master untuk konsistensi dan integritas data:

- **Customer Master**: Identitas unik pelanggan untuk pengalaman omnichannel.
- **Product Master**: Detail produk yang konsisten di semua kanal penjualan.
- **Supplier Master**: Informasi terpusat untuk pengadaan dan pembayaran.
- **Inventory Master :** Informasi terpusat tentang inventory yang tersedia.

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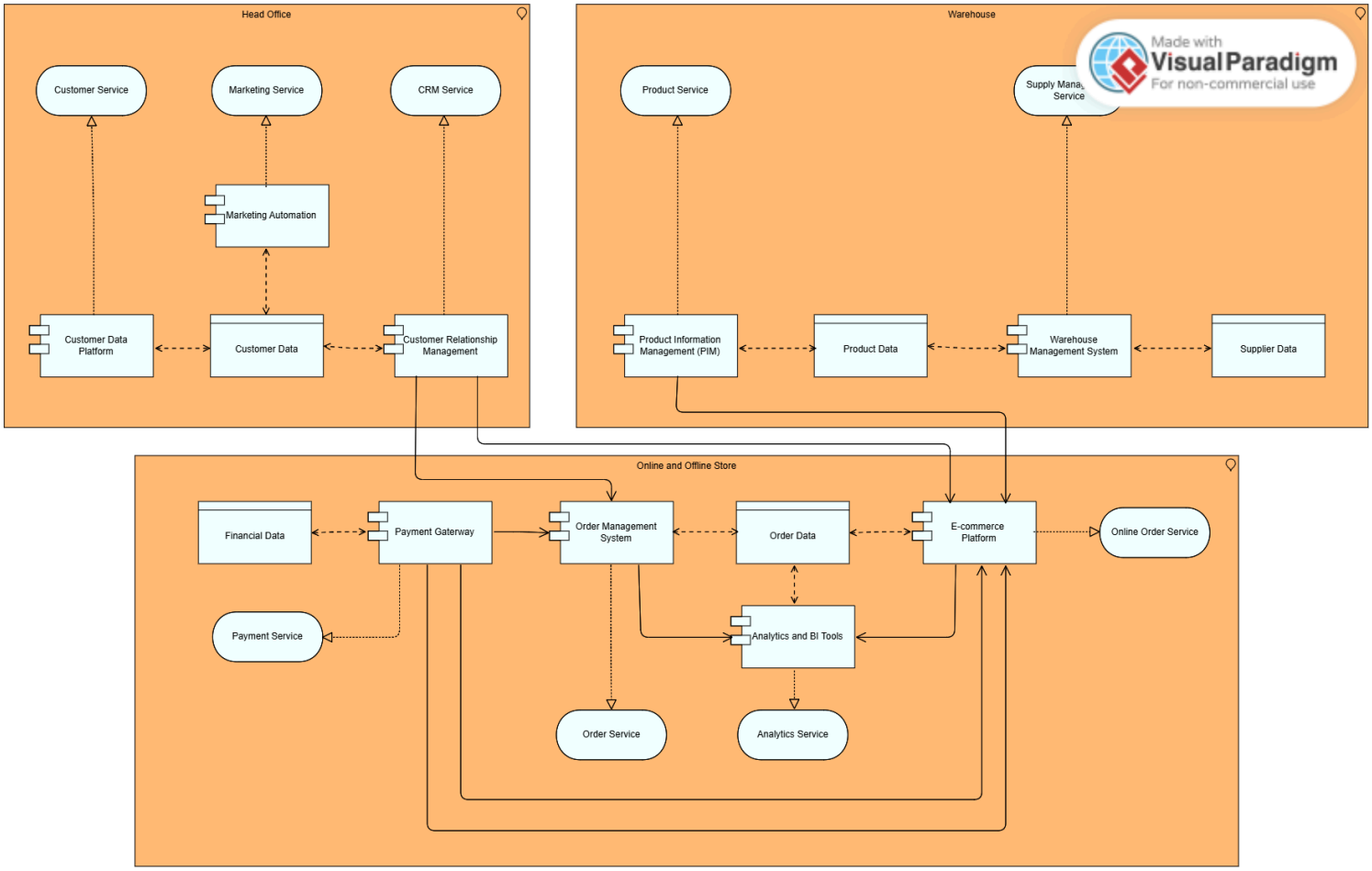
2. Application Architecture

Application Architecture menjelaskan aplikasi apa saja yang digunakan, bagaimana mereka berinteraksi, dan peran masing-masing aplikasi dalam mendukung proses bisnis.

A. Core Applications

1. **E-commerce Platform**



   - **Functionality**: Pencarian produk, katalog, keranjang belanja, checkout.
   - **Examples**: Magento, Shopify Plus, atau custom-built menggunakan React dan Node.js.
2. **Order Management System (OMS)**
   - **Functionality**: Pemrosesan pesanan, validasi pembayaran, pengiriman.
   - **Examples**: Salesforce Order Management, Oracle OMS.
3. **Warehouse Management System (WMS)**
   - **Functionality**: Manajemen inventaris, picking, packing, dan pengiriman.
   - **Examples**: Manhattan WMS, SAP Extended Warehouse Management.
4. **Customer Data Platform (CDP)**
   - **Functionality**: Mengumpulkan data pelanggan dari berbagai kanal.
   - **Examples**: Segment, Adobe Experience Platform.

5. **Payment Gateway**
    - **Functionality**: Otorisasi pembayaran, pengelolaan refund.
    - **Examples**: Shopee Pay, PayPal,Gopay, M Banking .

B. Supporting Applications

1. **Product Information Management (PIM)**
    - Mengelola katalog produk dan informasi atribut.
    - **Examples**: Akeneo, InRiver.
2. **Customer Relationship Management (CRM)**
    - Mengelola interaksi pelanggan, program loyalitas.
    - **Examples**: Salesforce CRM, HubSpot.
3. **Marketing Automation**
    - Email marketing, kampanye iklan, personalisasi.
    - **Examples**: Marketo, Klaviyo.
4. **Analytics and BI Tools**
    - Analisis data penjualan, pelanggan, dan inventaris.
    - **Examples**: Google Analytics, Tableau, Power BI.

C. Integration Architecture

1. **API Gateway**
    - Untuk integrasi antara aplikasi internal dan pihak ketiga.
    - **Examples**: Apigee, AWS API Gateway.
2. **Middleware and ESB (Enterprise Service Bus)**
    - Orkestrasi data dan penghubung antar aplikasi.
    - **Examples**: MuleSoft, Apache Camel.
3. **Event Streaming and Messaging**
    - Digunakan untuk real-time inventory update dan notifikasi pesanan.
    - **Examples**: Apache Kafka, RabbitMQ.

Technology Architecture

1. Infrastructure Architecture

Infrastructure Architecture menjelaskan komponen fisik dan virtual yang dibutuhkan untuk menjalankan sistem ritel secara skalabel dan handal.

A. Cloud Infrastructure

1. **Cloud Provider**:
    - **Public Cloud**: Menggunakan **AWS, Azure,** atau **Google Cloud Platform (GCP)**.
    - **Multi-Cloud Strategy** untuk redundansi dan fleksibilitas biaya.
2. **Compute Resources**:
    - **Virtual Machines (VMs)** untuk aplikasi legacy dan beban kerja intensif.
    - **Containerization** dengan **Kubernetes (EKS, AKS, GKE)** untuk aplikasi mikroservis.
3. **Storage Solutions**:
    - **Object Storage** untuk gambar produk dan file statis (AWS S3, Azure Blob).
    - **Block Storage** untuk database dan aplikasi kritis (EBS, Persistent Disk).
    - **Data Archival** untuk data historis (Glacier, Azure Archive).

B. Networking and Connectivity

1. **Virtual Private Cloud (VPC)**
    - Isolasi jaringan untuk keamanan dan skalabilitas
    - **Subnets**: Public untuk frontend, private untuk backend dan database.
2. **Content Delivery Network (CDN)**
    - **CloudFront, Azure CDN,** atau **Cloud CDN** untuk mempercepat loading website secara global.
3. **Load Balancing and Traffic Management**
    - **Application Load Balancer (ALB)** untuk distribusi beban aplikasi web.
    - **Global Traffic Manager** untuk multi-region failover.

4. **API Gateway dengan Zero Trust Model**
    - Untuk mengelola request API dengan autentikasi dan rate limiting.
    - **Examples**: **AWS API Gateway, Apigee, Kong.**

C. High Availability and Disaster Recovery

1. **Multi-Region Deployment**
    - Deploy aplikasi di beberapa region untuk redundansi dan latency rendah.
    - **Active-Active** atau **Active-Passive Failover**.
2. **Backup and Disaster Recovery**
    - **Snapshot Backup** untuk database dan VM.
    - **Cross-Region Replication** untuk object storage.
    - **Disaster Recovery Plan (DRP)** dengan **RPO** dan **RTO** yang jelas.

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2. Platform Architecture (Detail)

Platform Architecture mencakup platform dan layanan yang digunakan untuk pengembangan, penerapan, dan pengoperasian aplikasi.

A. Application Platform

1. **Container Orchestration**
    - **Kubernetes (EKS, AKS, GKE)** untuk mengelola container secara otomatis.
    - **Service Mesh** dengan **Istio** untuk observabilitas dan keamanan.
2. **Serverless Architecture**
    - **AWS Lambda, Azure Functions, Google Cloud Functions** untuk microservices event-driven.
3. **Middleware and Integration Platform**
    - **API Gateway** untuk expose dan mengelola API.
    - **Event Streaming** dengan **Apache Kafka** atau **AWS EventBridge**.

B. Database Platform

1. **Operational Databases**
    - **Relational Database**: **Amazon RDS, Azure SQL, Cloud SQL** untuk data transaksi.
    - **NoSQL Database**: **DynamoDB, Cosmos DB, Firestore** untuk data katalog dan pelanggan.
2. **Data Warehouse and Data Lake**
    - **Data Warehouse**: **Redshift, BigQuery, Synapse Analytics** untuk analitik bisnis.
    - **Data Lake**: **S3 Data Lake, Azure Data Lake** untuk data tidak terstruktur.
3. **Caching and In-Memory Storage**
    - **Redis atau Memcached** untuk caching session dan produk populer.

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3. Security Architecture (Detail)

Security Architecture menjelaskan langkah-langkah keamanan untuk melindungi data, aplikasi, dan infrastruktur.

A. Data Security

1. **Data Encryption**
    - **At-Rest Encryption** untuk database dan storage dengan **KMS (Key Management Service)**.
    - **In-Transit Encryption** menggunakan **TLS/SSL**.
2. **Data Masking and Tokenization**
    - Untuk melindungi data sensitif pelanggan dan pembayaran.

B. Application Security

1. **Identity and Access Management (IAM)**
    - **Role-Based Access Control (RBAC)** dan **Single Sign-On (SSO)**.
    - **Multi-Factor Authentication (MFA)** untuk pengguna internal dan pelanggan.
2. **Web Application Firewall (WAF)**
    - Untuk melindungi aplikasi dari serangan **DDoS** dan **OWASP Top 10**.

C. Network Security

1. **Virtual Private Network (VPN)**
    - **Site-to-Site VPN** untuk koneksi aman antara kantor dan cloud.
2. **Security Groups and NACLs (Network ACLs)**
    - **Zero Trust Architecture** dengan segmentasi jaringan

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4. Integration and Communication Architecture

1. **API Management**
    - **API Gateway** untuk routing, throttling, dan autentikasi API.
2. **Event-Driven Architecture**
    - Menggunakan **Apache Kafka, AWS EventBridge** untuk integrasi real-time.

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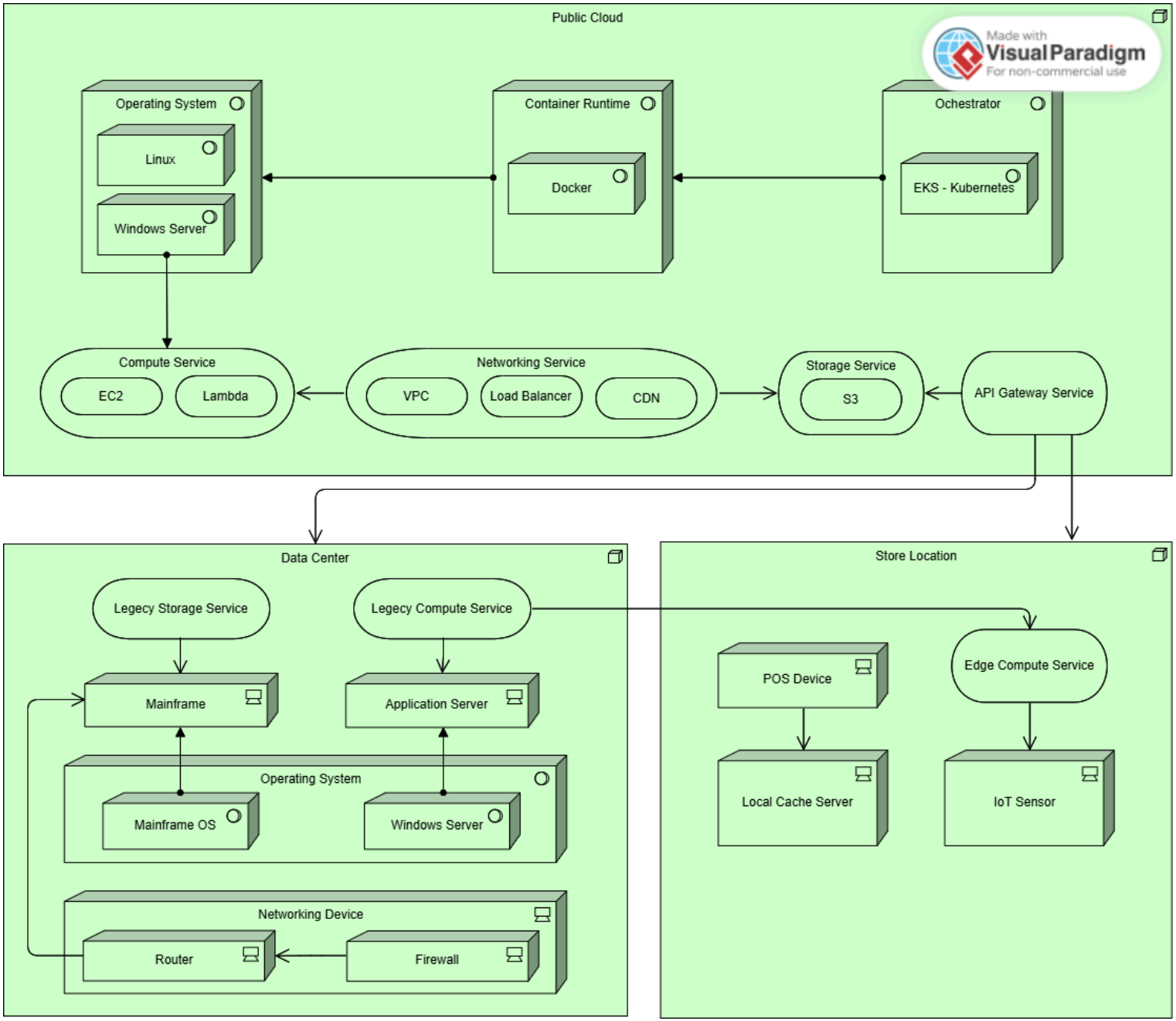
5. Deployment Architecture

1. **CI/CD Pipeline**
    - **Jenkins, GitHub Actions, GitLab CI/CD** untuk otomasi build dan deployment.
2. **Infrastructure as Code (IaC)**
    - **Terraform, AWS CloudFormation, Azure ARM Templates**.
3. **Deployment Strategy**
    - **Blue/Green Deployment, Canary Release, Rolling Update**


Public Cloud
Operating System
Linux
Windows Server
Container Runtime
Docker
Ochestrator
EKS - Kubernetes
Compute Service
EC2
Lambda
Networking Service
VPC
Load Balancer
CDN
Storage Service
S3
API Gateway Service
Data Center
Legecy Storage Service
Legecy Compute Service
Mainframe
Application Server
Operating System
Mainframe OS
Windows Server
Networking Device
Router
Firewall
Store Location
POS Device
Edge Compute Service
Local Cache Server
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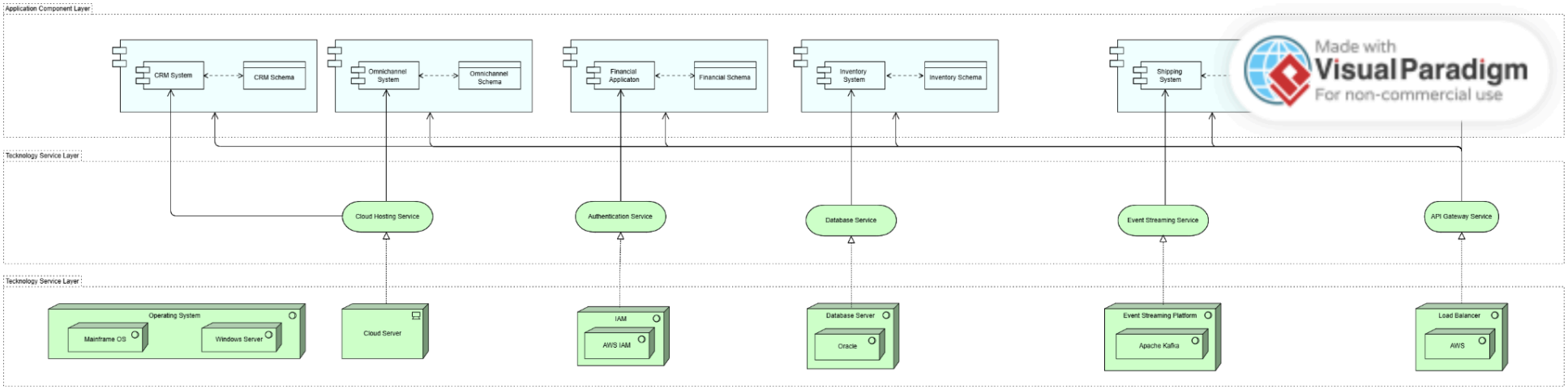

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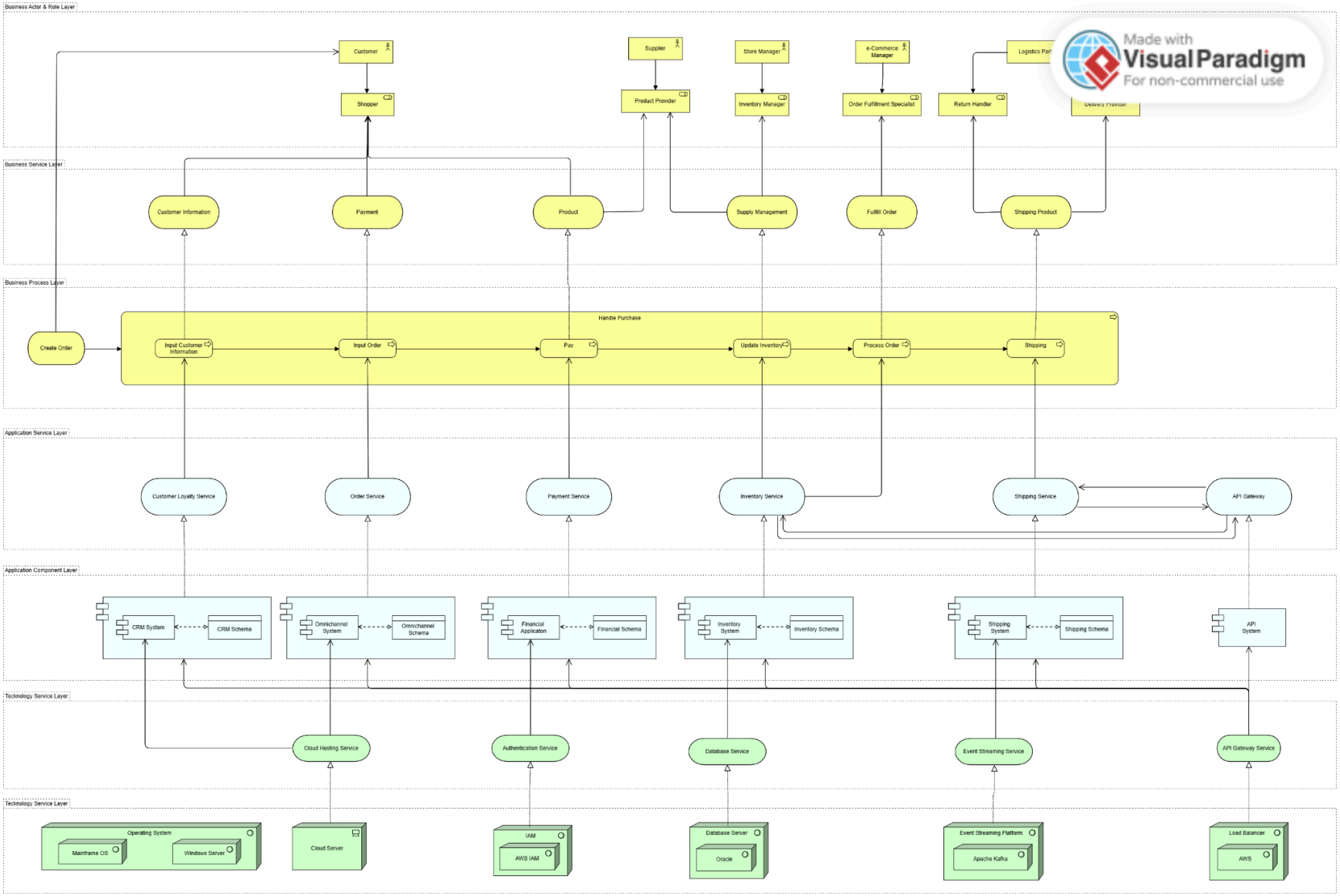
Integration Architecture



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